Task 3

3.1 Tool Name and Core Positioning

The jurisdiction-aware RegTech tool designed in this study is officially named FairLink: U.S.-Hong Kong Cross-Jurisdictional Fair Lending Compliance Tool, tailored specifically for The Bank of East Asia (Hong Kong). Its core positioning is one-click adaptation to U.S. and Hong Kong regulatory rules, automated algorithmic fairness detection, cross-jurisdictional compliance determination, explainable result output, lightweight and easy deployment (Hong Kong Compliance Tech, 2025), enabling rapid implementation without major modifications to BEA's existing risk control systems, effectively solving cross-jurisdictional credit compliance pain points, and helping BEA achieve compliant, fair, and efficient digital credit management (The Bank of East Asia, 2024).

3.2 Tool Prototype Architecture

The FairLink tool prototype adopts a modular, layered architecture design, consisting of 5 layers in total. Each layer has independent functions, clear interfaces, and works collaboratively, ensuring both concise and efficient core logic and good extensibility for adding new regulatory jurisdictions and expanding detection dimensions in the future (Hong Kong Compliance Tech, 2025).

(1) Data Input Layer: Responsible for receiving, cleaning, and standardizing BEA's raw credit application data. Supports three input methods: batch data import, real-time data interface, and manual data entry. Core input fields include customer credit score, gender, age, annual income (HKD), loan amount (HKD), occupation, loan purpose, and approval result. The data format is compatible with the output format of BEA's existing risk control systems, requiring no additional data conversion. Built-in data validation rules automatically identify missing values, outliers, and format errors, prompting users to correct them, ensuring input data quality (The Bank of East Asia, 2024).

(2) Jurisdiction Configuration Layer (Core Differentiation Layer): Built-in complete regulatory rule libraries for both the U.S. and Hong Kong. Rule parameters are configurable,

updatable, and version-manageable. Supports four core functions: one-click jurisdiction rule switching, custom rule parameters, batch rule updates, and rule change log tracking. The rule library contains key parameters such as gender approval rate difference threshold, age difference threshold, denial explanation requirements, compliance determination standards, and risk level classification (CPFB, 2026; HKMA, n.d.), fully matching the regulatory differences between the U.S. and Hong Kong. This is the core module enabling the tool's jurisdiction-aware capability.

(3) Fairness Detection Layer (Core Functionality Layer): Developed based on Python algorithms, achieving automated, multi-dimensional, high-precision algorithmic fairness detection (Hong Kong Compliance Tech, 2025). Core detection functions include calculation of gender group approval rate differences, age group stratified difference analysis, income distribution fairness validation, and overall compliance risk scoring. Built-in statistical analysis models automatically calculate key fairness indicators and generate quantitative detection results, avoiding manual calculation errors and accurately identifying hidden algorithmic biases.

(4) Explainable AI Layer: Automatically generates denial explanation reports that comply with local regulatory requirements based on jurisdiction rules. For the U.S. jurisdiction, generates detailed, specific, and easy-to-understand technical explanations, clearly listing the key factors affecting the denial (e.g., insufficient credit score, high debt-to-income ratio, insufficient repayment capacity, etc.) (CFPB, 2025). For the Hong Kong jurisdiction, generates concise, clear, non-technical brief explanations without excessive detail. The explanation reports support bilingual output (Chinese/English), PDF export, and one-click archiving, meeting regulatory disclosure and customer notification requirements (HKMA, 2025).

(5) Results Output Layer: Integrates fairness detection results, compliance determination conclusions, risk scores, denial explanation reports, and audit logs to generate visualized, easy-to-interpret compliance results reports. Output formats include written compliance

conclusions, quantitative indicator data, risk level labels (High/Medium/Low), detailed explanation reports, and audit logs. Supports real-time viewing, historical record query, result export, and print archiving, adapting to multiple usage scenarios such as compliance review, internal reporting, and regulatory audit for BEA (OCC, 2026).

3.3 Prototype Test Dataset: See GitHub

3.4 Prototype Code: See GitHub

3.5 Prototype Run Result Interpretation

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===== FairLink Prototype Test Results =====

[US Jurisdiction Results]
Jurisdiction: US
Compliance Standard: Strict Compliance
Male Approval Rate: 0.91
Female Approval Rate: 0.36
Gender Disparity (Absolute): 0.55
Compliance Result: Non-Compliant
Explanation Requirement: Detailed technical adverse action notice required

[HK Jurisdiction Results]
Jurisdiction: HK
Compliance Standard: Outcome-Based Compliance
Male Approval Rate: 0.91
Female Approval Rate: 0.36
Gender Disparity (Absolute): 0.55
Compliance Result: Non-Compliant
Explanation Requirement: No mandatory written explanation; brief verbal explan
ation if customer asks
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(1) Data Level: Among the 50 test data records, the male approval rate is 91%, the female approval rate is 36%, with a gender approval rate difference of 55%, indicating a gender fairness issue.

(2) U.S. Jurisdiction: The 55% difference exceeds the 10% rigid threshold, resulting in a determination of non-compliance, and a detailed technical denial explanation report is mandatorily required, fully matching the stringent U.S. regulatory requirements (CPFB, 2025, 2026).

(3) Hong Kong Jurisdiction: The 55% difference exceeds the 18% flexible threshold, resulting in a determination of non-compliance. No mandatory written denial explanation is required; only a brief explanation upon customer inquiry is needed, fully aligning with Hong Kong's outcome-oriented and flexible processing regulatory logic (HKMA, 2025).

(4) Core Value: The tool can precisely identify differences in compliance determination results and explanation requirements for the same data across different jurisdictions, perfectly realizing the core function of jurisdiction awareness and validating the effectiveness of the prototype design.

3.6 Model Card / Governance Documentation

The FairLink tool prototype is accompanied by a complete model card and governance documentation, clearly defining the tool's functions, assumptions, scope of application, failure modes, and risk prevention and control measures, meeting regulatory audit and internal governance requirements.

(1) Basic Tool Information

- Tool Name: FairLink U.S.-Hong Kong Cross-Jurisdictional Fair Lending Compliance Tool
- Tool Version: V1.0 (Prototype Version)
- Developer: Hong Kong Compliance Tech (HKCT)
- Applicable Institutions: The Bank of East Asia (Hong Kong) and similar multinational financial institutions
- Use Scenarios: Algorithmic fairness detection for personal credit approval, cross-jurisdictional compliance determination, denial explanation generation
- Core Functions: One-click jurisdiction rule switching, automated gender fairness detection, compliance result determination, differentiated denial explanation generation

(2) Core Assumptions

- Input data is authentic, valid, complete, and unaltered, with data distribution reflecting the actual characteristics of local customers in Hong Kong and the U.S.

- Gender is a core protected characteristic, and gender approval rate differences can effectively reflect the level of algorithmic fairness.
- Regulatory rule parameters for the U.S. and Hong Kong (difference thresholds, explanation requirements) are accurate and free from temporary policy changes.
- The algorithmic model is stable, with no malicious attacks, data contamination, or other anomalies.
- The manual review step is effective and can promptly correct tool misjudgments.

(3) Scope of Application

- Business Scope: BEA's personal consumer credit, residential mortgage loans, and credit card approval businesses.
- Jurisdiction Scope: Hong Kong, United States (extensible to Singapore, Australia, etc. in the future).
- Data Scale: Supports detection on small batches (tens of records) and medium batches (hundreds of records) of credit data.
- Compliance Dimensions: Prioritizes gender fairness coverage, extensible to age, race, geographic region, and other dimensions in the future.
- Excluded Scenarios: Corporate credit approval, cross-border data transmission, real-time high-frequency trading risk control, and non-credit financial services.

(4) Failure Modes and Risk Prevention & Control

- Failure Mode 1: Missing/Abnormal Data Leading to Detection Bias

Risk: Missing key fields, data format errors, or excessive outliers lead to incorrect fairness indicator calculations and compliance determination errors.

Prevention & Control: Built-in data validation rules automatically identify and prompt correction of missing/abnormal data; supports data cleaning and preprocessing functions to filter out invalid data.

- Failure Mode 2: Model Performance Drift Leading to Declined Fairness Detection Accuracy

Risk: Changes in market environment, customer characteristics, or data distribution lead to declined accuracy of the algorithmic detection model, making hidden biases difficult to identify.

Prevention & Control: Regularly update the model (quarterly) and retrain the algorithm; establish a model performance monitoring mechanism to provide real-time alerts for drift risks.

- Failure Mode 3: Jurisdiction Rule Updates Not Synchronized Leading to Invalid Compliance Determinations

Risk: Changes in U.S. or Hong Kong regulatory rules occur, but the tool's rule library is not updated in time, causing compliance determination standards to become outdated and resulting in misjudgments.

Prevention & Control: Establish a regular rule library update mechanism (monthly synchronization of regulatory updates); maintain rule change logs and version management; automatically trigger full-data re-detection after updates.

- Failure Mode 4: Human Operational Error Leading to Incorrect Jurisdiction Rule Configuration

Risk: Operational errors when manually switching jurisdiction rules cause incorrect loading of U.S. or Hong Kong rules, rendering compliance determinations completely invalid.

Prevention & Control: Built-in operation confirmation mechanism requiring double confirmation for rule switching; record all operation logs for traceability and auditability; set up rule configuration anomaly alerts.

- Failure Mode 5: Privacy Data Leakage Risk

Risk: When processing sensitive customer credit data, the tool faces risks of data leakage, unauthorized access, and misuse.

Prevention & Control: Encrypt data storage and transmission; implement strict access control, allowing only authorized personnel to access data; do not store raw sensitive data, only process anonymized data; comply with the requirements of Hong Kong's Personal Data (Privacy) Ordinance and the U.S. California Consumer Privacy Act (CCPA).

(5) Limitations

- The tool currently only covers gender fairness detection and does not yet include other protected characteristics such as age, race, or geographic region.

- Relies on input data quality; data distortion will directly affect the accuracy of detection results.
- Serves as a decision-support tool and is not a substitute for human review; final compliance determinations and credit approval decisions must be made by humans.
- Currently does not support real-time streaming data interface; data must be imported in batches for detection.
- The rule library requires manual periodic updates and cannot automatically identify regulatory policy changes.

References

Consumer Financial Protection Bureau. (2025). Guidance on AI-powered credit denial explanations.
https://www.consumerfinance.gov/about-us/newsroom/cfpb-issues-guidance-on-credit-denials-by-lenders-using-artificial-intelligence/

Consumer Financial Protection Bureau. (2026). Regulation B (12 CFR Part 1002).
https://files.consumerfinance.gov/f/documents/cfpb_reg-b_fr-notice-proposed-rule_2025-11.pdf

Hong Kong Compliance Tech. (2025). Company mission, core values, and business plan.
https://www.hkcompliancetech.com/company-overview

Hong Kong Monetary Authority. (2025). AI risk management guideline.
https://brdr.hkma.gov.hk/eng/doc-ldg/docId/getPdf/20241107-1-EN/20241107-1-EN.pdf

Hong Kong Monetary Authority. (n.d.). Code of banking practice for personal credit.
https://www.hkma.gov.hk/eng/key-information/guidelines-and-circulars/

Office of the Comptroller of the Currency. (2026). OCC Bulletin 2026-13: Artificial intelligence risk management. https://www.occ.treas.gov/news-issuances/bulletins/2026/bulletin-2026-13.html

The Bank of East Asia. (2024). Annual report 2024. https://www.hkbea.com/english/investor-relations/annual-reports.html